

INGESTION

Purpose: Handle the automated and manual ingestion of source data.

Main Responsibilities:

Function	Description
Extract Data	Pulls data from APIs, databases, file systems, etc.
Orchestrate Scheduling	Runs ETL jobs on a defined schedule (e.g., every hour, daily)
Incremental loads	Captures only new or updated records efficiently
Retry & Monitoring	Handles failures, retries, and sends alerts
Deliver to Staging Layer	Saves data to files, raw tables, or object storage

In the **data ingestion process**, various technologies are used depending on:

- **Where the data comes from** (source: database, file, API, etc.)
- **Where it's going** (destination: data warehouse, lake, DB, etc.)
- **How it's moved** (batch or real-time)

ways to perform data ingestion:

Batch Ingestion

- Data is collected and processed in **fixed intervals** (e.g., hourly, daily).
- Suitable for **large volumes** of data that don't require immediate processing.
- **Simpler to build and manage** compared to real-time systems.
- Common in reporting, analytics, and historical data loading.

List of open source technologies : There are different open source we need to select based on use case which is perfect fit.

Best Tools	Use cases
Apache Airflow	Flexible scheduling & complex workflows
Apache NiFi, Talend, Pentaho	Visual design with minimal coding
Apache Sqoop	Sync DBs to Hadoop
Meltano	Python-based ELT with connectors
Apache Airbyte	ingest data from apps/APIs to warehouse
Apache Beam	process and transform streaming or batch data at scale

NOTE: If you want one open-source tool that gives you **maximum flexibility, reliability, and scalability** for batch data ingestion — **Apache Airflow** is the top choice.

Apache Airflow – Key Points

- Workflow orchestration tool
- Built-in scheduling support
- Web UI for monitoring and logs
- Handles task dependencies
- Supports retries and alerts
- Highly extensible with custom plugins
- Works with files, APIs, databases, cloud, etc.
- Open-source and community-supported
- Used by companies like Airbnb, Netflix, Uber

Real-Time Ingestion

- Data is processed **immediately or within seconds** as it arrives.
- Suitable for **time-sensitive** applications like fraud detection, IoT, live dashboards.
- More **complex to design**, but offers **low latency**.
- Requires systems that support **streaming**

List of open sources to process real time injection: This are some of the sources we prefer the tool based on the requirement
There are few more sources ,just the mostly used tools are mentioned here.

Best tools	Use cases
Apache Kafka	Central messaging system for streaming
Apache Spark Streaming	Streaming pipelines with transformations
Apache Flink	Real-time analytics, fraud detection
Apache NiFi	Real-time data routing and transformation

NOTE: If you want one open-source tool that gives you **maximum flexibility, reliability, and scalability** for real-time/streaming data ingestion — Apache Kafka is the top choice.

Apache Kafka– Key Points

- **High Throughput:** Can handle **millions of messages per second**.
- **Durability:** Stores data on disk with configurable retention.
- **Scalability:** Easily scales horizontally by adding brokers.
- **Fault Tolerant:** Replication across multiple nodes ensures no data loss.
- **Flexible:** Supports multiple producers and consumers.
- **Widely Used:** Trusted by LinkedIn, Netflix, Uber, Airbnb, etc.

❖ list of tools that support both **batch and real-time ingestion**:

- Apache NiFi
- Apache Flink
- Apache Spark Structured Streaming
- Apache Beam
- Airbyte (Open Source)

